Introduction

Establishing referents is common in daily communication or stories. Incremental referential processing is a key mechanism that allows immediate integration of relevant context and linguistic information for identifying the intended referent (e.g. who the ‘nephew’ refers to when there were two nephews previously mentioned). A temporary confusion can be elicited if the discourse fails to support identification of the referent, which is defined as referential ambiguity. An event-related-potential (ERP) component called NRef effect is associated with the referential ambiguity in discourse processing, as indicated by a sustained frontal negativity shift in the electroencephalogram (EEG) right after the onset of the ambiguous noun phrase (Berkum et al., 1999 and Berkum et al., 2004).

Previous studies have shown that language processing is heavily context-based (Nieuwland & Berkum, 2006), e.g. through manipulating the feature of an inanimate character in the context, researchers found that participants’ later processing of the inanimate character would be more animate-alike. Same for referent processing, differences in context could reveal the cognitive mechanism of referential ambiguity. By altering elements within the story stimuli, Nieuwland et al. (2007) investigated referential ambiguity with the NRef component. The number of potential referent subjects were controlled by having one of the candidates left early in the context. The stimuli were given in three conditions: ambiguous (a nephew who was into history and another into politics both being present), unambiguous (a nephew who was into history being present and an uncle into politics left early), and disambiguated (a nephew who was into history being present and another into politics left early). The component window was the 2000 msec period after the critical sentence “he told the nephew…”. The result indicated that the NRef effect was found only in the ambiguous condition, not the disambiguated condition, with the unambiguous condition as the baseline for comparison. The result demonstrated that adults are utilizing a situation model to construct their understanding of the discourse. Adult participants' referential processing was successfully distinguished by the stimuli manipulation, as the mental processing of the situation or story model isolated the leaving character from the subsequent narration. A situation model of the discourse hinted that the ambiguity processing did not emerge from the level of text memory, but from a mental representation of the character dynamics (Zwaan & Radvansky, 1998).

How about the developmental trajectory of referential processing? Yuile and Fisher (2022) adopted the stimuli from Nieuwland et al., 2007 and modified them into child-friendly story versions to study referential ambiguity in young children. The story consisted of a play date between Maisy and her two animal friends (e.g. bunny with the crown and bunny/turtle with the necklace), and one of them left early. The critical sentence is Maisy asked the animal with the accessory to play something else. Instead of using EEG, Yuile and Fisher employed the approach of visual world paradigm (VWP) with an eye-tracker. VWP is a display of discourse-related contents on the screen for participants to freely search: four objects (in this study: target animal, competitor, and two play activities) were placed at each corner of a rectangle. The ambiguity measure was the preference look to the target over the competitor, and a larger target advantage (difference in look to target and competitor) presumed a lower ambiguity level and vice versa. The looking patterns observed in 4-5 year old participants mirrored the ambiguity processing of the ambiguous condition in the NRef study. However, a difference emerged as some level of ambiguity was found within the processing of disambiguated conditions. A comparable outcome was also observed within the adult comparison group. A possible explanation was that the target disadvantage in disambiguated condition was from the visual and linguistic distraction of the competitor in the VWP (Sikos et al., 2019). Semantic relatedness could introduce extra attention to phrases that are not explicitly mentioned but highly correlated to the mentioned ones. Within this framework, although the bunny with the necklace left early and was theoretically omitted from the situation model, the bunny with the necklace remained in the paradigm; thus the keyword “bunny” might shift attentional distraction to the bunny competitor, rather than toward a semantically-distinct turtle competitor. This explanation would only be possible if the same NRef effect pattern was observed within the new stimuli.

To investigate this question, we need to replicate the NRef effect pattern in Nieuwland et al before any further interpretation of the differences in outcomes between the two studies. The current study aims to conduct the adult version of the Yuile and Fisher study in ERP methodology.

Python Project Inspiration

The current python project is inspired by the feature of the NRef component defined across literature. Compared to other ERP components such as N400, P3 etc which have a clear window of interest, NRef is generally defined as a sustained negativity shift, the onset of the component and how long the sustain are neither investigated. Statistical significance of NRef in condition comparison was found in window 100-1500, 200-600, 600-1200 post stimulus onset across different studies (Lee & Lai, 2024; Nieuwland et al., 2007; Sikos et al., 2019). The variability in observed time windows may stem from the inherent complexity of semantic processing, which involves interactive mechanisms such as perception along with the encoding, decoding, and retrieval of memory (Ralph et al., 2017). The nature of EEG gives the data low spatial resolution but high temporal resolution, despite the high temporal resolution is from the compression of hidden components of a cognitive activity, e.g. the signal from a single electrode is a dipole triggered by multiple neural activities.

The sustained frontal negativity shift revealed significant semantic processing differences under the discourse manipulation. However, the lack of a defined temporal window obscured the boundary between semantic processing and simultaneous cognitive functions. Carrasco et al (2024) presented a multivariate pattern analysis approach of ERP data, which is implementing linear support vector machines (SVMs) to decode conditions. Their outcome shows that such an approach produced larger effect sizes than traditional univariate analysis. To run the SVM, the dataset was structured into a three-dimensional matrix after averaging individual raw trials into mini-ERPs: the rows correspond to each mini-ERP, the columns represent electrode sites, and the depth signifies the temporal dimension. They executed one SVM with cross-validation for every individual time point, which is 100 independent SVMs for 100 intervals of data. The SVM was trained with 80% of the data in the matrix and left the 20% as the test group. The SVM aimed to find the best hyperplane that separated the test data into conditions. The outcome of the analysis was a decoding accuracy of each time point, inferring the proportion correct for test phases. Their rationale for a linear SVM is that the weight vector provides an interpretable model for ERP analysis. The model assigned different weights to each feature (electrode in this case) with the training set, allowing the researchers to create topography out of the weight. Trammela et al (2023) also compared SVM, linear discriminant analysis, and random forest and concluded that SVM outperformed the other two models.

Although the initiative for SVM analysis is to address the statistical power crisis in ERP studies, performing individual SVMs at every time point facilitated a series of independent temporal decodings. Therefore, we can utilize the SVM approach to understand the discourse processing at a finer level. There is individual variability in semantic processing and thus shown at individual ERP waves. On the one hand, we can study what causes individual variance in the NRef time window, for example reading habits. On the other hand, the variation across studies mentioned in the previous paragraph might be induced by study design or stimuli instead of semantic processing.

The current project is to create a linear SVM analysis script for ERP data. The goal is to successfully run SVMs and generate decoding accuracy graphs for individuals and the group, including a report of statistical significance across time windows. The script is written based on the .set data from eeglab, a common file format for ERP studies. There was no cleaned data for NRef components from Carrasco et al (2024) thus the project testing runs on an open N400 data source (Swartz Center for Computational Neuroscience [SCCN], n.d.).

Data description

The 5-subject data for testing in the project were acquired by Peter Ullsberger and colleagues. The task was to distinguish between synonymous and non-synonymous word pairs (the second word presented 1 second after the first). Consequently, the primary investigation was the cognitive processing differences between these synonymous and non-synonymous word pairs. The ERP component of interest is N400, a negative brain wave peaking at roughly 400 ms post stimulus. This component serves as a direct indicator of language comprehension and semantic processing, typically presenting as a robust N400 effect when encountering incongruent linguistic input.

Design

The study is a replication of the Yuile and Fisher (2022) using an ERP framework. Participants will be fitted with a size-appropriate EEG net while conducting a behavior task. During the tasks, participants will passively listen to stories, no other tasks required in progress. Participants will be seated in front of a monitor screen where visual displays will be presented. At the beginning of each story, a picture of the protagonist will appear at the center of the screen; A rectangular arrangement will feature two animal companions and two activities positioned at its four vertices, as shown in figure 1. As the story progresses to the critical sentence, the two animal friends will disappear from the screen, while the picture of the protagonist remains visible. The image of the protagonist serves as the attention cue to help maintain participants' gaze at the center of the screen. Comprehension questions will be asked at the end of the story for attention check. Across the three experimental conditions, participants will be randomly presented with a total of 72 stories, with 24 stories allocated to each condition. The entire procedure is organized into three distinct blocks, each comprising 24 stories. Participants will be provided with a brief rest period between these blocks. Scalp EEG will be measured continuously throughout the task.

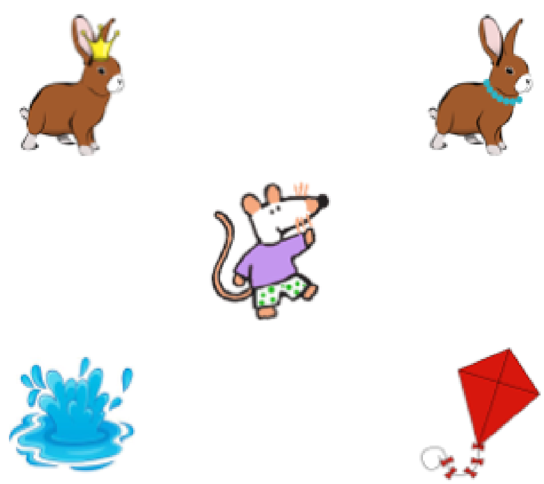

Figure 1. Display example

Story Example

Introduction	One day Maisy was at the beach. She was with her friends the bunny (1-1: turtle) with the necklace and the bunny with the crown. Maisy was wearing her polka dot shorts.	
Conditions	2-2 (ambiguous):	The bunny with the necklace was flying a kite and so was the bunny with the crown.
	2-1 (disambiguated):	The bunny with the necklace had to leave early, but the bunny with the crown was flying a kite.
	1-1 (unambiguous):	The turtle with the necklace had to leave early, but the bunny with the crown was flying a kite.
Critical	Maisy didn't like flying kites; she liked splashing in the water. *She asked* ***the bunny with the*** *crown to splash in the water with her*. And then they splashed in the water together.	

SVM analysis

The best decoding outcome of guessing is 50% because the SVM matrix incorporates an equal distribution of each condition. The statistical analysis is a one-way t-test for testing whether the decoding accuracy is significantly better than pure chance .5 (H0: decoding accuracy = .5). To enhance interpretability, decoding accuracy was averaged over 100-ms intervals rather than compared at every individual millisecond window.

Predicted Results

The ERP result should show a similar pattern of NRef as Nieuwland et al., (2007), as in figure 2. Taking the unambiguous condition wave as a baseline, the NRef effect should be observed only in ambiguous condition, not in disambiguated condition. Alternatively, if the disambiguated condition is processed similarly to the ambiguous one, a reduced NRef effect might be observed. Another possibility is a lack of NRef effects across both conditions, which would suggest the stimuli manipulation was unsuccessful..

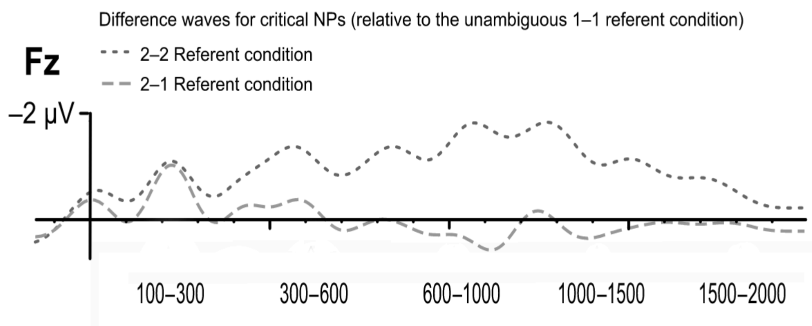

Figure 2. Difference waves from Nieuwland et al for critical noun phrases.

SVM decoding result

In terms of SVM decoding accuracy, an increasing and sustained high decoding accuracy is expected after a certain time window, probably after 600 msec. Based on the current study design where a visual change (all images disappear except the protagonist one) occurs a short period before the critical sentence, I also expect an early window of high decoding accuracy due to the visual processing.

The preliminary results from the SVM decoding analysis, as illustrated in figure 3, demonstrate that the 300-400 ms time window yielded the highest accuracy relative to other intervals, which is the commonly defined window for the N400 component. While this initial testing was performed using the 5-subject N400 dataset rather than NRef data, the findings suggest that linear SVM decoding is a valid and effective approach for conducting temporal analysis on ERP signals.

In terms of the NRef outcome analysis, if the decoding accuracy fails to present a clearer time window, it tells us that the NRef component represents a more complicated cognitive mechanism that creating this scene of various time windows all contributes to the decoding process. Thus more experimental design can be adopted to distinguish different necessities for a psycholinguistic task like this. Conversely, identifying a distinct temporal window via decoding accuracy would imply that traditional univariate methods may obscure the temporal characteristics of ERP signals.

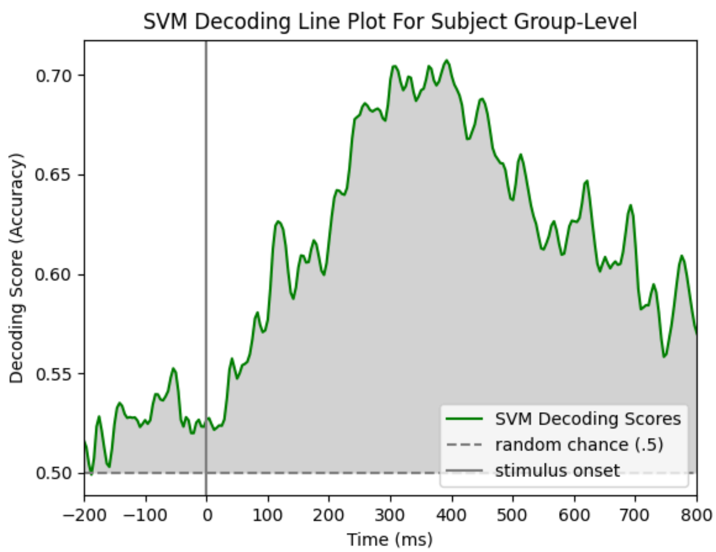

Figure 3. SVM decoding accuracy for the 5-subject N400 data

Discussion

Based on the previous literature and experimental design, we hypothesized that the NRef effect will emerge exclusively when comparing the ambiguous condition to the unambiguous condition, as the departure of a single referent should resolve the initial ambiguity in disambiguated condition. Our study diverges from previous research Nieuwland et al., by adapting child-friendly narrative stimuli for use within the VWP and utilizing illustrated stories rather than speech-only presentation.

This methodological shift is critical because it addresses whether the previously observed elimination of target advantage in the VWP is a result of referential processing or merely a byproduct of visual distraction. Specifically, in disambiguated trials, the presence of two on-screen images sharing the same name and conceptual category may induce attentional interference, a confound also noted by Sikos et al. (2019) regarding eye-movement patterns. By conducting this direct comparison, we aim to disentangle cognitive referential processing from attentional distraction, thereby refining the utility of these two methods for investigating the neural and behavioral signatures of language comprehension.

The SVM decoding would offer a valuable insight into the temporal dynamics and duration of the NRef effect, in this case clarifying whether the visual processing resides beyond the stimulus onset. Furthermore, it could also serve as a practical measure to explore how task manipulation affects the brain waves at different time windows.

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